The Recommended Analysis of the Biomarker-by-Drug Interaction in Hybrid N-of-1 Trials: Expectancy Terms, a Working Covariance and CR2 Inference, with Worked Examples

2026-10-09 09:41 PDT

Author. pmsimstats team

Purpose. The exploratory stress test of `docs/46` and `docs/47` compared fourteen analyses of the biomarker-by-drug interaction $\beta_{bm:D}$ in the Hybrid design and selected one of them, called F1. This note explains that analysis term by term: what each part of the model is for, what goes wrong without it, and what it costs. Two worked examples carry the explanation. The first fits F1 and the simpler analysis S to one simulated trial, with every coefficient, standard error and degree of freedom shown. The second reads the stored stress-test results to show the same contrast over thousands of trials. A short section explains why choosing the carryover half-life by AIC, as the variant F1A did, fails the size criterion, and what the project intends instead; the full treatment of a data-chosen half-life is in the companion note `docs/57`. The note ends with a recipe in R. It is written for trial statisticians and methodologists who will use the analysis, including the authors of the original pmsimstats code [1, 2], and can be read without the other project documents.

Contents

Notation index and glossary . . . 4
1. Summary for the busy reader . . . 5
2. Scope . . . 6
3. The model, term by term . . . 6
3.1 The centered biomarker and the time adjustment . . . 6
3.2 $D_{bc,it}$ and the pre-exposure guard . . . 6
3.3 The expectancy terms η_t and $B_i^c\eta_t$ . . . 7
3.4 Random intercept plus CAR(1): a working covariance . . . 9
3.5 CR2 standard errors with Satterthwaite degrees of freedom . . . 9
4. Worked example 1: one trial, step by step . . . 10
4.1 The trial . . . 10
4.2 The coefficients . . . 10
4.3 The test of the interaction . . . 11
4.4 A large trial: the bias made visible . . . 11
4.5 The fitted slopes by visit type . . . 12
5. Worked example 2: the stored Monte Carlo results . . . 12
5.1 Size . . . 12
5.2 Size-adjusted power . . . 13
6. The analysis half-life . . . 15
6.1 What F1A did, and why it failed size . . . 15
6.2 Why a fixed half-life is right-sized even when it is wrong . . . 16
6.3 Status and intention . . . 16
7. Alternatives and when to prefer them . . . 18
8. Practical recipe . . . 19
9. Limitations . . . 20
10. Reproducibility . . . 20
11. References . . . 21

List of Tables

1 Notation index for this note . . . 4
2 The analyses discussed in this note (`docs/47`, notation table; inspected in the driver) . . . 5
3 $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week on the four Hybrid paths (verified) . . . 7
4 Biomarker slope fitted by S and F1 in each group of rows, with $D_{bc,it}$ near 1 on drug and near 0 off drug (derived; `docs/49`, Section 3) . . . 8

5 Worked example 1, no interaction ($c_{bm} = 0$, prognostic natural course): coefficients of S and F1 with model-based and CR2 standard errors (verified) . . . 11

6 Worked example 1: the interaction test under S and F1, model-based and CR2 inference, seed 20261009 (verified) . . . 11

7 Worked example 1 at $N = 2000$: the interaction estimate against its target (verified) . . . 12

8 Null rejection rates by cell and analysis, stress test as run on 2026-10-07 (verified; MCSE 0.0069 per cell, 0.0021 pooled; criteria: pooled within 0.046 to 0.054, every cell at most 0.075) 12

9 Size-adjusted power at $c_{bm} = 0.30$ (0.25 in the CS curve), $N = 70$ unless stated, 500 trials per cell (verified, `decision-power-ds-nosym-all14.csv`; MCSE at most 0.022). Values in parentheses belong to analyses whose test is biased in that cell; size adjustment corrects the level, not the bias . . . 13

10 F1A against F1 on the same null trials of the 11 standard cells, by the half-life F1A selected (verified) . . . 16

11 Alternatives to F1 (verified from the stored stress-test outputs; prognostic column: the largest of the four prognostic null cells) . . . 18

List of Figures

1 Worked example 1: slope of Y on the biomarker by visit type, implied by the S fit (solid, triangles) and the F1 fit (dashed, circles), with the true slope of the construct (dotted, squares) and the slope observed within each visit group of the simulated trial (crosses). Top: the $N = 70$ trial; bottom: the $N = 2000$ trial. Left: no interaction; right: $c_{bm} = 0.30$. The biomarker is prognostic through the natural course, so its true slope is 0 at baseline and -0.195 at every later visit, plus the predictive part on drug. S must use one slope for baseline and the blinded off-drug visits; it compromises between them, and the on-drug visits absorb the rest as an interaction. F1 can bend at baseline and between open-label and blinded visits. . . . 14

2 Worked example 2. Left: null rejection rates in the 15 null cells for S, F1, ES, F1A and R; the shaded band is 0.05 ± 2 MCSE for one cell, the dashed line the cap of 0.075, and the dotted vertical line separates the 11 standard cells from the four prognostic ones. R's rate in the placebo cell, 0.159, is drawn at the top edge. Right: size-adjusted power along the AR(1) curve ($N = 70$; the stored curve has no null point, so lines join $c_{bm} > 0$ only, and $c_{bm} = -0.30$ is shown as an isolated point). . . . 15

3 AIC selection of the analysis half-life on six null trials. Top: ML AIC minus its minimum over the half-life grid, for F1 with the interaction (solid; the F1A rule) and without it (dashed; the proposed F1A0 rule), true half-life 0.5 week (left) and 1 week (right). The dotted line marks 2 AIC units. Bottom: the CR2 t-statistic of the interaction in the same trials across the grid; filled circles mark the half-life chosen by the F1 AIC, open circles that chosen by the F1A0 AIC. Colors identify trials. . . . 17

Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`). Symbols marked *local* are defined here for this note only.

Table 1: Notation index for this note

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t (code `Sx`); $t = 0$ is baseline; higher is worse	canonical
BL_i, TV_{it}, PB_{it}, BR_{it}	baseline level; natural-course, placebo (expectancy) and drug responses; $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$	canonical
B_i	biomarker of participant i (code `bm`)	canonical
B_i^c	biomarker centered at the trial's baseline mean, $B_i - \bar{B}$, not scaled (code `bmc`)	local
D_{it}	binary drug state at visit t	canonical
$D_{bc,it}$	exposure-decayed drug indicator: 1 on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, 0 at baseline and before first exposure (code `Dbc`; the stress-test driver calls it `X`)	canonical
t_{sd}	time since discontinuation, in weeks (code `tsd`); 0 on drug and before first exposure	canonical
$t_{1/2}$	true carryover half-life of the data-generating process	canonical
$t^a_{1/2}$	half-life assumed by the analysis in $D_{bc,it}$	canonical
η_t	expectancy: 0 at baseline, 1 in the open-label phase, 0.5 when blinded (code `De`)	canonical
β_B	biomarker main effect (coefficient of `bmc`)	local
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand (coefficient of `bmc:Dbc`)	canonical
β_η, $\beta_{bm:\eta}$	coefficients of `De` and `bmc:De`	local
u_i	participant random intercept	canonical
ρ_a	working CAR(1) correlation per week	local
c_{bm}	biomarker-drug-response coupling (the predictive effect)	canonical
$c_{bm,TV}$	biomarker-natural-course coupling (a prognostic effect)	local
s_{prog}	prognostic slope of Y on B that does not depend on drug	local (`docs/48`)
f	share of s_{prog} absorbed by a shared β_B	local (`docs/48`)
w_0, w_1, w_2, w_3	total weights (row counts) of baseline, open-label on-drug, blinded on-drug and blinded off-drug rows	local
κ	mean standard error over the empirical SD of the estimate	canonical
ν	denominator degrees of freedom	canonical
N	total sample size across paths	canonical

Glossary.

- **Hybrid design.** Eight visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 after a baseline at week 0: open-label treatment at weeks 4 and 8, a blinded discontinuation, and a brief crossover, over four randomization paths [1]. Every open-label visit is on drug.
- **Predictive biomarker.** Changes the size of the drug effect; this is what $\beta_{bm:D}$ measures.
- **Prognostic biomarker.** Associated with the symptom score whatever the treatment, for example through the natural course or the placebo response.
- **Working covariance.** The covariance structure an analysis assumes for a participant's repeated scores in order to weight them. It need not be the true covariance.

- **CR2.** The bias-reduced cluster-robust (sandwich) variance estimator of Bell and McCaffrey [3], clustered on participant, with the Satterthwaite-type degrees of freedom of Pustejovsky and Tipton [4].
- **Size.** The false-positive rate of the two-sided test of $\beta_{bm:D} = 0$ at $\alpha = 0.05$.
- **Size-adjusted power.** Power when each analysis rejects at the 5% quantile of its own p-values in the matching null cell, so that analyses are compared at equal size.
- **MCSE.** Monte Carlo standard error, $\sqrt{p(1-p)/R}$ for a rate p over R trials [10]: 0.0069 at $p = 0.05$ with $R = 1{,}000$, at most 0.022 with $R = 500$.
- **The analyses.** All are fitted to the long data with baseline as a response row at $t = 0$, linear time in weeks, and $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week unless stated.

Table 2: The analyses discussed in this note (`docs/47`, notation table; inspected in the driver)

ID	Mean model	Working covariance	Inference
S	`Sx ~ bmc + t + Dbc + bmc:Dbc`	u_i + CAR(1)	CR2
F1	S + `De + bmc:De`	u_i + CAR(1)	CR2
F1A	F1, $t^a_{1/2}$ chosen by AIC over {0, 0.25, 0.5, 1, 2}	u_i + CAR(1)	CR2
ES	S + `bmc:bl0` (`bl0` = 1 at baseline)	u_i + CAR(1)	CR2
G1, G2	F1	u_i + CAR(1), residual SD by phase	CR2 (G1), model-based (G2)
DS	S, $t^a_{1/2}$ chosen by AIC	u_i + CAR(1)	CR2
R	the published-type reference: S with binary D_{it}	compound symmetry	model-based, Satterthwaite

Evidence labels. Each claim carries one of six labels: *quoted* (reproduced from a source), *inspected* (read from source code or a stored file), *derived* (follows algebraically), *verified* (computed by running code; the script is named in Section 10), *argued* (reasoned but not computed), or *proposal* (a procedure not yet run).

Status of the stored results. Every Monte Carlo number in this note is from the stress test as run on 2026-10-07. That run used the published recursive carryover on the drug-response mean and a fixed allocation (18, 18, 17, 17 at $N = 70$), in configuration C at $\rho = 0.7$ (`docs/47`, Section 2). The driver now defaults to anchored carryover and randomized allocation; a rerun under those defaults is pending. The worked examples of Sections 4 and 6 use the same recursive, fixed-allocation construct so that they match the stored results (inspected).

1. Summary for the busy reader

The model. For participant i at visit t (baseline included),

$$Y_{it} = \beta_0 + \beta_B B^c_i + \beta_T t + \beta_D D_{bc,it} + \beta_{bm:D}\, B^c_i D_{bc,it} + \beta_\eta \eta_t + \beta_{bm:\eta}\, B^c_i \eta_t + u_i + e_{it},$$
$$u_i \sim N(0, \sigma^2_u), \qquad \mathrm{Cor}(e_{it}, e_{is}) = \rho_a^{|t-s|} \text{ (weeks)},$$

fitted by REML with `nlme::lme`, and the test of $\beta_{bm:D} = 0$ made with the CR2 cluster-robust standard error and its Satterthwaite degrees of freedom (`clubSandwich::coef_test`). Here $D_{bc,it}$ uses a half-life $t^a_{1/2}$ fixed in advance from pharmacology (0.5 week in the stress test) and is 0 before any exposure.

When to use it. For a Hybrid or similar N-of-1 design whose open-label phase is all on drug, whenever the biomarker might also be prognostic, which for a physiological marker such as blood pressure cannot usually be excluded (`docs/49`, Section 6).

What it protects against (verified, stored stress test).

- *A prognostic biomarker.* With the biomarker correlated 0.3 with the natural course or with the placebo response and no interaction, S rejects 0.099 and 0.100 and the published-type analysis R 0.107 and 0.159; F1 rejects 0.047 and 0.040.
- *A misspecified working covariance.* Over 11 standard null cells F1 rejects 0.051 pooled, and at most 0.062 in any cell, including the cells where the true half-life (0 or 1 week) differs from the assumed 0.5 week. R, with a compound-symmetry working covariance and model-based inference, rejects 0.069 pooled and 0.081 at $N = 35$.

What it costs. Power. At $c_{bm} = 0.30$, $N = 70$, under AR(1) data, F1's size-adjusted power is 0.364 against 0.514 for S, a relative loss of 29%; over ten AR(1) alternative cells the loss ranges from 9% to 30%. Under compound-symmetry data it is small (0.522 against 0.554) (verified). Plan the sample size for F1, not for S.

The one thing not to do. Do not choose the half-life by AIC and then test as if it had been fixed. That is F1A, and it fails the size criterion: 0.057 pooled and 0.073 when the true half-life is 1 week (verified; Section 6).

Status. The validated analysis is F1 with a half-life fixed in advance. The project's intended analysis lets the data choose the half-life, by a procedure that does not reuse the interaction's noise (the proposal F1A0) or that calibrates the test for the selection (the proposal F1AB). Neither has been evaluated yet; both are set out in `docs/57`. Until they are, report F1 at the fixed half-life, with a sensitivity analysis over a grid of half-lives and the data's own estimate of the half-life reported alongside (Section 8).

2. Scope

This note covers one analysis family for one estimand in one design family: the biomarker-by-drug interaction in aggregated Hybrid N-of-1 trials with a Gaussian outcome. It does not reopen the selection; the rule, the cells and the amendments are in `docs/46` and `docs/47`. It explains the selected analysis and gives worked examples. Where an argument is approximate, it says so.

3. The model, term by term

3.1 The centered biomarker and the time adjustment

Centering. The biomarker enters as $B_i^c = B_i - \bar{B}$, centered at the trial's baseline mean and not scaled (inspected, driver `make_trial()`). Centering makes β_D the drug effect at an average biomarker and keeps the biomarker block separate from the rest of the model: because B_i^c has mean zero and is independent of the randomized path, its cross-products with the intercept, time and $D_{bc,it}$ vanish in the limit, and the interaction's probability limit depends only on the biomarker columns (derived; `analysis/report/compendium-mathematics.tex`, eq:bp-plim). Scaling changes nothing that is tested: t-statistics and p-values are invariant to rescaling B_i^c, and $\hat{\beta}_{bm:D}$ is then per unit of the raw biomarker (inspected; `docs/47`, note below the notation table). In this note's construct $\sigma_{bm} = 15.36$, so a slope of -0.195 per unit is -3.0 points per SD.

Time. Linear time in weeks absorbs the average drift of the natural course and the placebo response. It enters only the non-biomarker block, so it does not change the interaction's probability limit, and `docs/42` found estimates that agree to within 0.0009 across time adjustments (quoted there; derived in the compendium, eq:bp-plim). A richer time adjustment is a matter of efficiency, not of bias, for $\beta_{bm:D}$.

3.2 $D_{bc,it}$ and the pre-exposure guard

What it does. The drug effect does not vanish on the day the drug is stopped. $D_{bc,it}$ codes exposure as 1 on drug and $2^{-t_{sd}/t^a_{1/2}}$ afterwards, so the first blinded off-drug visit still counts as partly exposed; carryover in within-participant designs, and the cost of ignoring it, are long-standing concerns [9]. When $t^a_{1/2}$ equals

the true half-life and the biomarker coupling decays the same way, the true biomarker slopes lie exactly in the span of $(1, D_{bc,it})$ and the interaction is recovered in full under any working covariance (derived; compendium, eq:invf-dbc-exact). With $t^a_{1/2} = 0$ it reduces to the binary D_{it}.

Table 3 shows the coding on the four Hybrid paths at $t^a_{1/2} = 0.5$ week (verified).

Table 3: $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week on the four Hybrid paths (verified)

Week	η_t	Path 1	Path 2	Path 3	Path 4
0 (baseline)	0	0	0	0	0
4	1	1	1	1	1
8	1	1	1	1	1
9	0.5	1	1	1	1
10	0.5	1	1	0.25	0.25
11	0.5	0.25	0.25	0.0625	0.0625
12	0.5	0.0625	0.0625	0.0156	0.0156
16	0.5	1	0.0002	1	0.0001
20	0.5	0.0039	1	0.0039	1

The guard. The time since discontinuation is 0 both on drug and before the first exposure, and $2^0 = 1$. A decay function written as `ifelse(on_drug, 1, 0.5^(tsd / t_half))` therefore codes a participant who starts on placebo as fully exposed at every placebo visit. In the Hybrid design every path starts on drug, so the guard never acts there (inspected, Table 3); it matters in any design with a placebo-first sequence. On the placebo-first path of the two-period crossover (eight visits 2.5 weeks apart, drug from week 12.5) at $t^a_{1/2} = 1$ week, the guarded indicator is 0, 0, 0, 0, 1, 1, 1, 1 and the unguarded one is 1 at all eight visits (verified). The unguarded participant then has no within-participant variation in exposure and contributes nothing to $\beta_{bm:D}$. The project found this in its own carryover manuscript: in a paired rerun of its crossover cells ($N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week, 500 replicates), restoring the guard raised the power of the exposure-weighted analysis from 0.512 to 0.814, the level of the binary indicator, 0.810 (quoted; report 01, Appendix B.4). The guarded form, as in the driver and in the patched `exposuredecay()` (inspected):

```
exposure_decay <- function(on_drug, tsd, t_half) {
  if (t_half <= 0) return(as.numeric(on_drug))
  ifelse(on_drug, 1, ifelse(tsd > 0, 0.5^(tsd / t_half), 0))
}
```

3.3 The expectancy terms η_t and $B^c_i\eta_t$

This is the part of F1 that the other analyses lack, and the reason it was selected.

The confounding the Hybrid design builds in. Every open-label visit is on drug and carries full expectancy, $\eta_t = 1$; every blinded visit has $\eta_t = 0.5$; baseline has neither. Drug and expectancy therefore move together. Over the nine rows at the stress-test allocation, the allocation-weighted correlation of η_t with the binary D_{it} is 0.588 (0.611 with $D_{bc,it}$; 0.507 over the eight post-baseline rows) (verified; it reproduces the compendium's 0.588). Anything that scales with expectancy looks, on average, like something that scales with drug.

What a prognostic biomarker does to a shared slope. Suppose the biomarker predicts the natural course but not the drug effect, so the true slope of Y on B is 0 at baseline and s_{prog} at every later visit, on drug or off. S has two biomarker slopes: β_B for every row coded off drug, baseline included, and $\beta_B + \beta_{bm:D}$ on drug. It must fit one β_B to the baseline rows (true slope 0) and the off-drug rows (true slope s_{prog}), so $\hat{\beta}_B$ settles between them, $\hat{\beta}_B \approx f s_{\text{prog}}$. The on-drug rows have slope s_{prog}, and the only parameter left to close the gap is the interaction:

$$\hat{\beta}_{bm:D} \approx (1-f)\, s_{\text{prog}} \neq 0$$

(derived; `docs/48`, Section 2; compendium, eq:bp-leak). Under working independence with binary D, $1-f = w_0/(w_0+w_3)$, the weight share of the baseline rows among the rows coded off drug. In the Hybrid design at the stress-test allocation the row counts are $w_0 = 70$, $w_1 = 140$, $w_2 = 176$ and $w_3 = 244$ (verified), so $1-f = 0.223$. A correlated working covariance changes the factor: the compendium computes 0.276 for S's random-intercept CAR(1) fit in the published construct, and 0.629 in configuration C under the true covariance (quoted; compendium, tab:bp-leak and tab:invd-leakC). The bias is fixed, not random, so the false-positive rate grows with N. That a baseline response row imposes such a constancy constraint is known from the literature on baseline as a covariate or a response [8].

What the expectancy terms do. F1 adds η_t and $B_i^c \eta_t$. The biomarker slope the model can now fit is different in each of the four groups of rows:

Table 4: Biomarker slope fitted by S and F1 in each group of rows, with $D_{bc,it}$ near 1 on drug and near 0 off drug (derived; `docs/49`, Section 3)

Rows	S	F1
baseline ($\eta = 0$)	β_B	β_B
open label, on drug ($\eta = 1$)	$\beta_B + \beta_{bm:D}$	$\beta_B + \beta_{bm:\eta} + \beta_{bm:D}$
blinded, on drug ($\eta = 0.5$)	$\beta_B + \beta_{bm:D}$	$\beta_B + 0.5\beta_{bm:\eta} + \beta_{bm:D}$
blinded, off drug ($\eta = 0.5$)	β_B	$\beta_B + 0.5\beta_{bm:\eta}$

Two things follow (derived; compendium, Section "Freeing the baseline slope, and the expectancy terms").

1. *Baseline is freed.* Because $\eta_0 = 0$, the baseline slope is no longer tied to the blinded off-drug slope; $\beta_{bm:\eta}$ absorbs the difference.
2. *An expectancy-scaled prognostic slope is absorbed exactly.* If the biomarker acts through the placebo response, its slope is proportional to η_t, which lies in the span of F1's biomarker columns, so the leak is zero under any working covariance.

The interaction is then identified mainly from the comparison the Hybrid design was built to make clean: blinded on drug against blinded off drug, at the same expectancy.

Candidly: F1 is approximately, not exactly, unbiased here. For a prognostic slope that is constant after baseline (the natural-course pathway), F1's fitted slopes obey one linear restriction, and the weighted projection under working independence with binary D gives

$$\hat{\beta}_{bm:D} \to s_{\text{prog}} \frac{1/w_3 - 1/w_2}{1/w_0 + 1/w_1 + 1/w_2 + 1/w_3}$$

(derived; compendium, eq:bp-f1-leak). The leak vanishes only when the blinded on-drug and off-drug rows carry equal weight, $w_2 = w_3$. In the Hybrid design $w_2 = 176$ and $w_3 = 244$, so the factor is -0.051 under independence with binary D, -0.020 with D_{bc}, 0.089 under a random-intercept CAR(1) working covariance and 0.171 under the true AR(1) covariance in the published construct, and 0.162 under the true covariance of configuration C (quoted; compendium, tab:bp-leak and tab:invd-leakC). These factors are a fifth to a third of S's under the same covariances, which is why F1's size stayed at 0.047 in that cell; but the protection is quantitative, not exact.

ES has the mirror-image weakness. ES frees baseline directly with `bmc:bl0` and is exact for any pathway constant after baseline, but an expectancy-scaled slope is larger at the open-label visits, which are all on drug, and leaks into ES's interaction by 0.14 to 0.24 of s_{prog} depending on the working covariance (quoted; compendium, tab:bp-leak). Its stress-test size in the placebo cell, 0.049, shows the leak was small at a coupling of 0.3 (verified), and its mean estimate there, -0.027, against F1's -0.001, shows it was not zero (verified, `summary-full.csv`).

3.4 Random intercept plus CAR(1): a working covariance

The random intercept u_i and the continuous-time AR(1) residual, `corCAR1(form = ~ t | ptID)` on the week scale, are a *working* covariance: a parsimonious device, three parameters (σ_u^2, σ_e^2, ρ_a), for weighting a participant's nine scores sensibly. It is not a claim that the data have this structure, and in the stress-test construct they do not. The true outcome covariance has three phase-specific variance levels (341.6 at baseline, 709.6 open label, 598.6 blinded), no noise of its own at baseline, and cross-phase correlations the structure cannot reproduce exactly (quoted; compendium, Section "The implied outcome covariance and phase-specific variances"). A random intercept absorbs only the long-lag part of AR(1) correlation (compendium, eq:invari). The working model need only give reasonable weights; the correctness of the standard error is delegated to CR2.

Richer working covariances bought no robustness in the stress test, and some made the prognostic leak worse. Residual variances by phase (C1, C2) up-weight the baseline row, whose variance is smallest, and S's leak factor rises toward 0.63; C1 rejected 0.382 in the natural-course prognostic cell (quoted; compendium, tab:invd-leakC; verified, `decision-screen-ds-nosym-all14.csv`). Once F1's expectancy terms free the baseline slope, the same phase variances are harmless (G1 0.055, G2 0.062 in that cell) but add little power (Section 7).

3.5 CR2 standard errors with Satterthwaite degrees of freedom

The intuition. A model-based standard error trusts the working covariance: it computes the variance of $\hat{\beta}$ as if the data had exactly that structure. The sandwich estimator [5] does not. It keeps the working model's weights (the bread) but measures the variability of each participant's contribution to the estimating equation (the meat) from the observed residuals, participant by participant,

$$\widehat{\mathrm{Var}}(\hat{\beta}) = \Big(\sum_i X_i' \hat{V}_i^{-1} X_i\Big)^{-1} \Big(\sum_i X_i' \hat{V}_i^{-1} \hat{e}_i \hat{e}_i' \hat{V}_i^{-1} X_i\Big) \Big(\sum_i X_i' \hat{V}_i^{-1} X_i\Big)^{-1}$$

(quoted; compendium, eq:sandwich). Because participants are independent, the observed spread of their contributions is a consistent estimate of the true variance whatever the working covariance. A wrong working covariance makes the estimate less efficient; it does not make the standard error wrong.

Why CR2 and not the plain sandwich. Residuals are smaller than the errors they estimate, because each participant's own data pulled the fit toward them; with 35 to 70 participants the plain sandwich is noticeably too small. CR2 inflates each participant's residuals by $(I - H_{ii})^{-1/2}$, with H_{ii} that participant's block of the hat matrix, which makes the estimator exactly unbiased when the working model is right and nearly so when it is not [3] (compendium, eq:cr2).

Why the degrees of freedom matter. The precision of a sandwich standard error is governed by the number of participants and by how unevenly they inform the coefficient, not by the number of rows. For $\beta_{bm:D}$, a participant-level biomarker times a within-participant exposure contrast, a few participants with extreme biomarker values carry much of the information. The Satterthwaite approximation of Pustejovsky and Tipton [4] matches the first two moments of the CR2 variance to a scaled χ^2 and returns $\nu = (\sum_g \ell_g)^2 / \sum_g \ell_g^2$ (compendium, eq:cr2-df). In the worked example of Section 4 it is 25.8 to 30.5 at $N = 70$, against 555 to 557 for the `nlme` containment degrees of freedom of the model-based test (verified); over five verification data sets the mean CR2 degrees of freedom were 21.2 for S and 20.5 for F1 (quoted; patch README, "Verification results"). At $\nu \approx 25$ the two-sided critical value is 2.06 rather than 1.96, so the correction matters at the sample sizes these trials use.

The evidence. Three pieces bear on it.

- *The published-type analysis is liberal.* R (compound symmetry, model-based standard errors, Satterthwaite degrees of freedom, binary D) rejected 0.069 pooled over the 11 standard null cells, 0.081 at $N = 35$, and its κ was 0.88 to 0.98; the CR2 analyses S and F1 rejected 0.048 and 0.051 (verified,

Table 8 and `56-kappa.csv`). R differs from S in working covariance and exposure coding as well as in inference, so this comparison does not isolate the inference.

- *The clean comparison is `docs/45`.* There the same compound-symmetry working covariance was fitted with model-based and with CR2 inference: mean size 0.079 against 0.051 over eight null cells, and 0.158 against 0.060 in the crossover under AR(1) data, with κ 0.73 against 0.95 (quoted; `docs/45`, Table 4.1). Kenward-Roger inference [6] did not repair it (0.079), because it corrects for estimating a covariance assumed correct, not for a wrong one (quoted; `docs/45`, Section 4.2).
- *Where the working covariance is close, model-based inference is also fine.* With phase-specific residual variances, which approach the stress-test construct's covariance, model-based and CR2 inference on the same fits gave pooled sizes of 0.052 and 0.053 (C2, C1) and 0.047 and 0.049 (G2, G1) (verified, `decision-screen-ds-nosym-all14.csv`). CR2 is recommended because a trial's analyst cannot know in advance that the working covariance is close.

4. Worked example 1: one trial, step by step

4.1 The trial

One Hybrid trial of $N = 70$ (18, 18, 17, 17 per path) is drawn from the stress-test cell `prog-TV-const` with the driver's own `build()`, `draw()` and `make_trial()`: configuration C at $\rho = 0.7$, $t_{1/2} = 0.5$ week, recursive carryover, and the biomarker correlated $c_{bm,TV} = 0.3$ with the natural-course component at every visit (inspected). The biomarker is therefore prognostic. The true prognostic slope is

$$s_{\text{prog}} = -\frac{c_{bm,TV}\,\sigma_{TV}}{\sigma_{bm}} = -\frac{0.3 \times 10}{15.36} = -0.195$$

points per unit of biomarker after baseline and 0 at baseline (derived; the minus sign because $Y = \text{BL} - (TV + \cdots)$). The first version has no interaction, $c_{bm} = 0$. The second uses the same seed, 20261009, with $c_{bm} = 0.30$, for which the true interaction is $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm} = -0.30 \times 8/15.36 = -0.156$ (derived). The two draws share the seed but not the covariance, so they are not the same participants shifted by an effect; they are two trials from the same random stream.

Loading the driver's functions without running the driver:

```
source('analysis/scripts/recommended-analysis/56-load-driver.R')
env <- load_driver(carryover = 'recursive', allocation = 'fixed')
cl <- env$cell('prog-TV-const', 0, c_tv = 0.3)     # c_bm = 0
st <- env$cell_setup(cl)
set.seed(20261009L)
d <- env$make_trial(cl, st$bs, env$alloc(cl, st), 1)
dd <- env$with_x(d, 0.5)                            # X = D_bc at 0.5 week
fit_S  <- env$fit_lme(env$F_S, dd)                  # Sx ~ bmc + t + X + bmc:X
fit_F1 <- env$fit_lme(env$F_F1, dd)                 # + De + bmc:De
clubSandwich::coef_test(fit_F1, vcov = 'CR2', test = 'Satterthwaite')
```

The fits are the driver's `fit_lme()` with its `lmeControl`, so they are exactly the stress-test analyses (inspected; the script checks that the driver's `cr2()` returns the same standard error). Both fits together took 0.4 to 1.2 seconds (verified).

4.2 The coefficients

Table 5: Worked example 1, no interaction ($c_{bm} = 0$, prognostic natural course): coefficients of S and F1 with model-based and CR2 standard errors (verified)

Term	S estimate	S SE, model / CR2	F1 estimate	F1 SE, model / CR2
intercept	78.42	2.98 / 2.77	79.11	3.06 / 2.59
`bmc` (β_B)	−0.159	0.193 / 0.183	−0.171	0.200 / 0.169
`t`	−0.528	0.119 / 0.090	−0.514	0.119 / 0.094
`Dbc` (β_D)	−7.50	1.22 / 1.37	−6.83	1.38 / 1.42
`De` (β_η)			−2.21	2.11 / 1.93
`bmc:De` ($\beta_{bm:\eta}$)			0.030	0.147 / 0.116
`bmc:Dbc` ($\beta_{bm:D}$)	−0.040	0.085 / 0.085	−0.050	0.097 / 0.100

The drug lowers the score by about 7 points on drug, and the time slope is about half a point per week of improvement. The biomarker terms are all small relative to their standard errors.

4.3 The test of the interaction

Table 6: Worked example 1: the interaction test under S and F1, model-based and CR2 inference, seed 20261009 (verified)

Case	Analysis	$\hat{\beta}_{bm:D}$	SE (model)	ν (model)	p (model)	SE (CR2)	ν (CR2)	p (CR2)
$c_{bm} = 0$	S	−0.040	0.085	557	0.636	0.085	30.5	0.637
$c_{bm} = 0$	F1	−0.050	0.097	555	0.607	0.100	30.5	0.623
$c_{bm} = 0.30$	S	−0.296	0.097	557	0.0024	0.072	25.8	0.0004
$c_{bm} = 0.30$	F1	−0.224	0.108	555	0.039	0.088	25.8	0.017

Reading.

- *With no interaction, neither analysis rejects*, and in this trial S shows no visible spurious interaction: its estimate, −0.040, is half a standard error from zero, and F1's is similar. This is the typical case. S's bias, $(1 - f)s_{\text{prog}}$, is a fraction of a standard error at $N = 70$, so a single trial cannot reveal it; it shows up as a doubled false-positive rate over many trials (Section 5) and as a clear offset in a large trial (Section 4.4).
- *With the interaction, both reject at 0.05.* F1's estimate is closer to the true −0.156 and has the larger standard error (0.088 against 0.072 under CR2); that standard error is the price of not trusting the baseline row (Section 3.3).
- *Model-based and CR2 standard errors differ in both directions*, and the degrees of freedom differ twentyfold. In the trial with the interaction CR2 gives the smaller standard error; over many trials the two are close on average for these analyses (κ near 0.94 to 1.03; Section 5), and CR2's value is that it remains right when the working covariance is wrong.
- $\hat{\beta}_{bm:\eta}$, the coefficient of `bmc:De`, is near zero without the interaction (0.030) and −0.241 (CR2 SE 0.172) with it: in a single trial it also picks up part of the predictive signal, because the open-label visits are on drug (verified, `56-example-coefficients.csv`).

4.4 A large trial: the bias made visible

The same construct at $N = 2000$ (500 per path, seed 20261010) shows what the $N = 70$ trial hides (verified; fits took 170 seconds).

Table 7: Worked example 1 at $N = 2000$: the interaction estimate against its target (verified)

Case	Analysis	$\widehat{\beta}_{bm:D}$	CR2 SE	$\widehat{\beta}_{bm:D}/s_{\text{prog}}$	Target
$c_{bm} = 0$	S	−0.0565	0.0150	0.289	0
$c_{bm} = 0$	F1	−0.0184	0.0170	0.094	0
$c_{bm} = 0.30$	S	−0.1908	0.0140		−0.156
$c_{bm} = 0.30$	F1	−0.1464	0.0156		−0.156

With no interaction, S reports an interaction of −0.057, 3.8 CR2 standard errors from zero ($p = 0.0002$), and its ratio to the prognostic slope, 0.289, is close to the leak factor computed for S's working covariance in the published construct, 0.276 (quoted; compendium, tab:bp-leak). F1's estimate, −0.018, is 1.1 standard errors from zero, and its ratio, 0.094, is close to that computed for F1, 0.089: the small residual bias of Section 3.3, visible at this size. With the interaction, S overshoots the target by −0.035, about the size of its null leak, while F1 is within one standard error. The 1,000 stored null trials of this cell at $N = 70$ agree: mean estimates −0.051 for S, −0.017 for F1 and 0.002 for ES (verified, `summary-full.csv`, cell 25).

4.5 The fitted slopes by visit type

Figure 1 plots, for each group of rows, the biomarker slope implied by each fit, averaged over the rows of the group, against the true slope of the construct and the slope observed in the trial.

The bottom-left panel is the mechanism in one picture (verified). The true slope is 0 at baseline and −0.195 after. S fits −0.129 at baseline and −0.134 at the blinded off-drug visits (one shared slope, moved slightly by the carryover in $D_{bc,it}$) and −0.185 on drug, so the difference, its interaction, is −0.051 per row-average (the coefficient, −0.057, applies to the full on-drug contrast). F1 fits −0.082 at baseline, −0.224 open label, −0.162 blinded on drug and −0.146 blinded off drug: it still misfits each group, but the blinded on-drug and off-drug slopes differ by only −0.017.

5. Worked example 2: the stored Monte Carlo results

5.1 Size

Table 8 gives the null rejection rates of six analyses in each of the 15 null cells, 1,000 trials each, read from the stored summaries (verified; `summary-full.csv`, `summary-full-extra.csv`, `summary-full-extra2.csv`; pooled rates agree with `decision-screen-ds-nosym-all14.csv` to 3×10^{-5}).

Table 8: Null rejection rates by cell and analysis, stress test as run on 2026-10-07 (verified; MCSE 0.0069 per cell, 0.0021 pooled; criteria: pooled within 0.046 to 0.054, every cell at most 0.075)

Null cell	S	F1	ES	F1A	DS	R
AR(1) curve	0.049	0.062	0.059	0.068	0.054	0.076
CS curve	0.048	0.057	0.048	0.069	0.050	0.050
measurement error	0.051	0.047	0.041	0.057	0.053	0.071
$N = 35$	0.058	0.053	0.050	0.059	0.058	0.081
$N = 150$	0.047	0.047	0.050	0.047	0.050	0.063
$t_{1/2} = 0$	0.036	0.049	0.042	0.043	0.040	0.072
$t_{1/2} = 1$ week	0.043	0.061	0.050	0.073	0.055	0.061
Weibull, shape 0.5	0.047	0.046	0.047	0.058	0.054	0.073
Weibull, shape 2	0.037	0.045	0.044	0.040	0.040	0.066
20% MCAR dropout	0.059	0.051	0.049	0.064	0.068	0.079
20% MAR dropout	0.056	0.044	0.046	0.051	0.059	0.064

Null cell	S	F1	ES	F1A	DS	R
pooled, 11 standard cells	**0.048**	**0.051**	**0.048**	**0.057**	**0.053**	**0.069**
prognostic: natural course, constant	0.099	0.047	0.049	0.059	0.115	0.107
prognostic: natural course, growing	0.049	0.046	0.063	0.052	0.058	0.053
prognostic: placebo response	0.100	0.040	0.049	0.047	0.112	0.159
prognostic: baseline severity	0.050	0.040	0.050	0.045	0.055	0.075

Reading.

- *Only the analyses that free the baseline slope survive the prognostic cells.* S, DS and R double their false-positive rate when the biomarker tracks the natural course or the placebo response; F1 and ES do not. A baseline-severity association enters every visit equally and harms no analysis materially (R reaches the cap, 0.075).
- *F1's largest cell is 0.062*, the AR(1) curve's null point, within the cap and about 1.7 MCSE above 0.05. Its κ there is 0.94 (verified, `56-kappa.csv`): CR2 is slightly small at $N = 70$ in this cell, as the small-sample theory allows.
- *F1 is right-sized when its half-life is wrong.* With the true half-life 0 or 1 week and $t^a_{1/2}$ fixed at 0.5, F1 rejects 0.049 and 0.061; under Weibull decay 0.046 and 0.045 (Section 6.2).

5.2 Size-adjusted power

Table 9: Size-adjusted power at $c_{bm} = 0.30$ (0.25 in the CS curve), $N = 70$ unless stated, 500 trials per cell (verified, `decision-power-ds-nosym-all14.csv`; MCSE at most 0.022). Values in parentheses belong to analyses whose test is biased in that cell; size adjustment corrects the level, not the bias

Alternative cell	S	F1	ES	F1A	G1	G2	DS	R
AR(1) curve, $c_{bm} = 0.30$	0.514	0.364	0.364	0.338	0.354	0.380	0.486	0.398
CS curve, $c_{bm} = 0.25$	0.554	0.522	0.662	0.446	0.546	0.528	0.480	0.596
$N = 35$	0.222	0.192	0.206	0.218	0.214	0.218	0.232	0.208
$N = 150$	0.828	0.748	0.704	0.748	0.752	0.754	0.822	0.734
$t_{1/2} = 0$	0.594	0.496	0.504	0.522	0.510	0.532	0.588	0.496
$t_{1/2} = 1$ week	0.388	0.272	0.288	0.282	0.282	0.306	0.412	0.334
measurement error	0.400	0.318	0.336	0.296	0.320	0.330	0.402	0.382
20% MAR dropout	0.418	0.382	0.356	0.372	0.404	0.378	0.424	0.378
prognostic: natural course, constant	(0.606)	0.482	0.370	0.448	0.520	0.524	(0.590)	(0.522)
prognostic: natural course, growing	0.516	0.378	0.192	0.386	0.484	0.490	0.522	0.448
prognostic: placebo response	(0.652)	0.446	0.498	0.434	0.428	0.424	(0.610)	(0.590)

Reading.

- *The cost of F1 is real and depends on the serial correlation.* Under AR(1) data F1 loses 29% of S's power at the reference cell, and between 9% (MAR dropout) and 30% ($t_{1/2} = 1$ week) over ten AR(1) alternative cells (verified, `56-cost-F1-vs-S.csv`). Under compound symmetry the loss is 6% (0.522 against 0.554). The loss comes from no longer using the baseline row as a clean observation of the off-drug biomarker slope (`docs/45`, Section 6; `docs/48`, Section 1).
- *F1 keeps its power under every prognostic form*; ES loses most of it when the natural-course association grows over time (0.192).
- *G1 and G2 are effectively equivalent to F1* in the standard cells and somewhat better in the natural-course prognostic cells (0.520, 0.524 against 0.482; 0.484, 0.490 against 0.378).
- *None of the robust analyses reaches power 0.80 on the grid at $N = 70$*; at $N = 150$, F1 reaches 0.748.

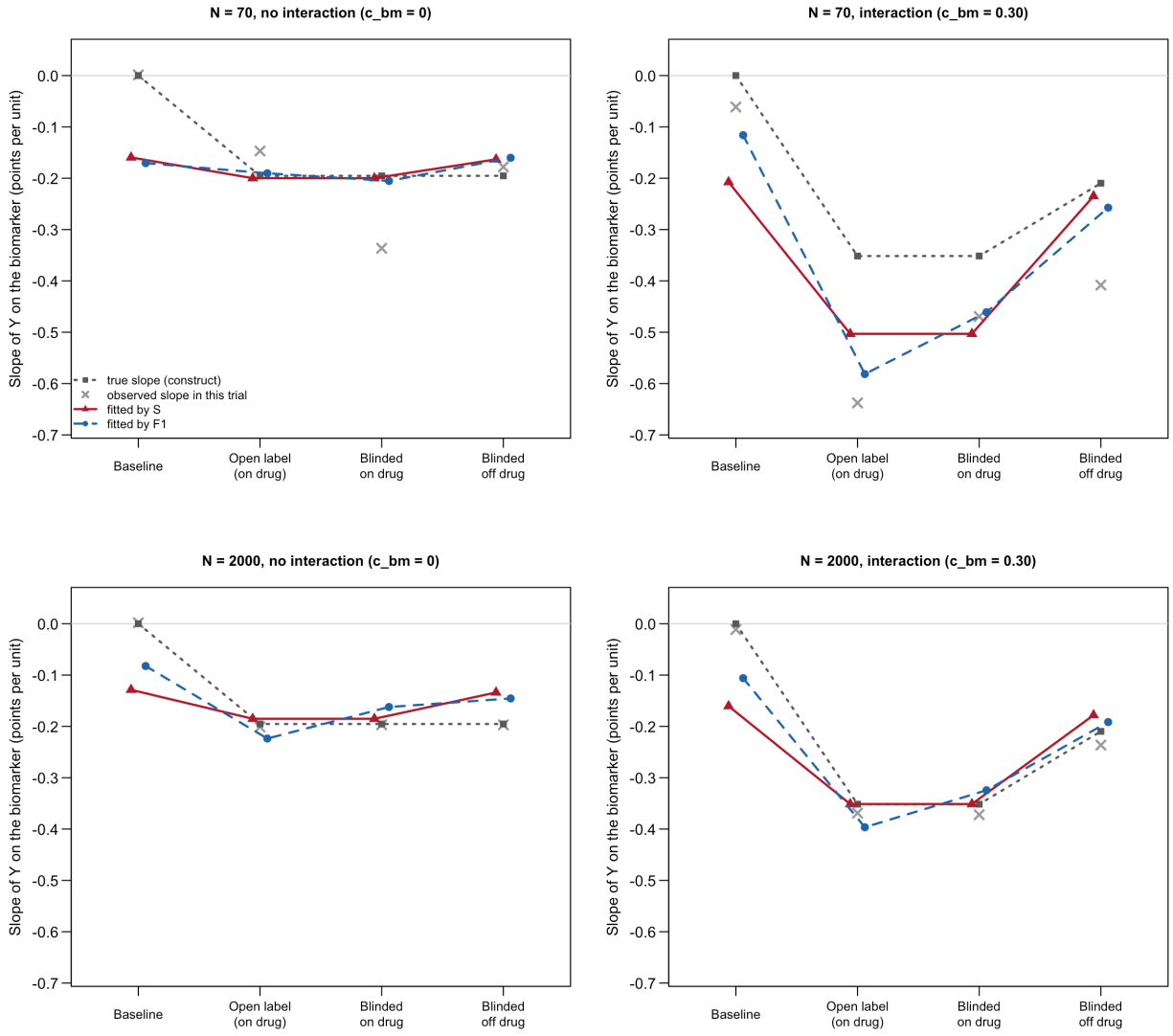


Figure 1: Worked example 1: slope of Y on the biomarker by visit type, implied by the S fit (solid, triangles) and the F1 fit (dashed, circles), with the true slope of the construct (dotted, squares) and the slope observed within each visit group of the simulated trial (crosses). Top: the $N = 70$ trial; bottom: the $N = 2000$ trial. Left: no interaction; right: $c_{bm} = 0.30$. The biomarker is prognostic through the natural course, so its true slope is 0 at baseline and -0.195 at every later visit, plus the predictive part on drug. S must use one slope for baseline and the blinded off-drug visits; it compromises between them, and the on-drug visits absorb the rest as an interaction. F1 can bend at baseline and between open-label and blinded visits.

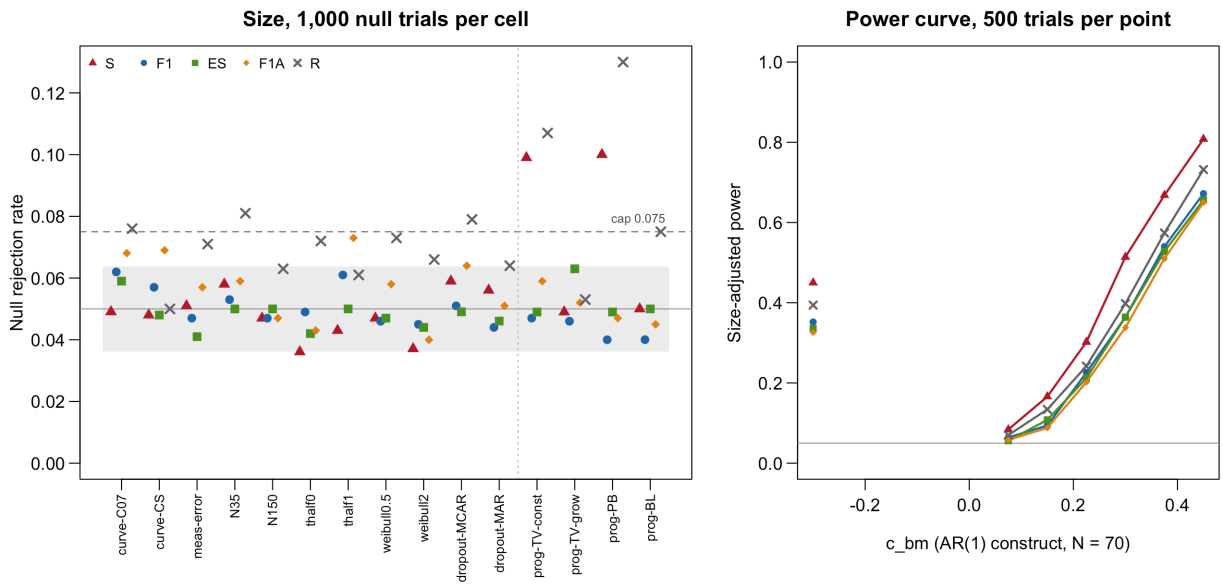


Figure 2: Worked example 2. Left: null rejection rates in the 15 null cells for S, F1, ES, F1A and R; the shaded band is 0.05 ± 2 MCSE for one cell, the dashed line the cap of 0.075, and the dotted vertical line separates the 11 standard cells from the four prognostic ones. R's rate in the placebo cell, 0.159, is drawn at the top edge. Right: size-adjusted power along the AR(1) curve ($N = 70$; the stored curve has no null point, so lines join $c_{bm} > 0$ only, and $c_{bm} = -0.30$ is shown as an isolated point).

6. The analysis half-life

The full treatment of choosing $t^a_{1/2}$ from the data is in `docs/57`. This section states only why the variant that was run failed, why a fixed half-life does not, and what the project intends.

6.1 What F1A did, and why it failed size

F1A fitted F1 by maximum likelihood at each $t^a_{1/2} \in \{0, 0.25, 0.5, 1, 2\}$ weeks, took the value with the smallest AIC, refitted that model by REML and tested $\beta_{bm:D}$ with CR2 as if the half-life had been fixed (inspected, driver `fit_extra2()`). Because the candidates differ only in the values of one regressor, they have the same number of parameters, and the AIC rule is the maximum-likelihood estimate of the half-life over the grid (derived; compendium, eq:invd-aic).

It is tempting because carryover is real and the data carry some information about it. It fails because the selection and the test use the same noise.

- *The likelihood includes the interaction term.* The half-life is chosen with `bmc:Dbc` in the model, so a value of $t^a_{1/2}$ that happens to line up the exposure regressor with chance biomarker structure in the drug contrast raises the likelihood and the t-statistic together (argued).
- *The test then ignores the choice.* The reported standard error is that of a model whose regressor was fixed, not estimated.

The stored paired results show the consequence (verified; `replicates-full.rds` and `replicates-full-extra2.rds`, the same trials, matched by cell and replicate; S agrees exactly between the two runs).

Table 10: F1A against F1 on the same null trials of the 11 standard cells, by the half-life F1A selected (verified)

Selected $t^a_{1/2}$	Trials	F1 size (fixed 0.5)	F1A size	Mean $\|t\|$, F1	Mean $\|t\|$, F1A
0	3,148	0.051	0.053	0.812	0.848
0.25	1,655	0.048	0.051	0.823	0.830
0.5	2,934	0.052	0.052	0.839	0.839
1	1,967	0.056	0.064	0.842	0.858
2	1,295	0.046	0.079	0.807	0.915
all	10,999	0.051	0.057	0.826	0.853

When AIC moves the half-life away from 0.5 week, and especially to 2 weeks, the test rejects more often than F1 on the same trials. Over the 10,999 trials, 135 are rejected by F1A only and 68 by F1 only. In the null cell with true half-life 1 week, 77% of selections were 1 or 2 weeks and F1A rejected 0.073 (0.083 when it chose 2 weeks), against F1's 0.061 (verified, `56-f1a-thalf1-by-h.csv`).

How well AIC identifies the half-life. Coarsely. Over all cells the selections were spread 24%, 16%, 29%, 19% and 12% over 0, 0.25, 0.5, 1 and 2 weeks (verified; as in `docs/47`, Section 8), but that pools cells with different true half-lives. By cell, AIC picked 0 in 87% of the null trials when the true half-life was 0, and 1 or 2 weeks in 77% when it was 1 week, but the true 0.5 week in only 38% of the null trials of the base cell, and 1 or 2 weeks in 85% of the compound-symmetry curve's null trials, whose true half-life is also 0.5 week (verified, `56-stress-summary.R` output). The likelihood distinguishes "little carryover" from "a week or more", not 0.25 from 0.5 or 1, and its choice depends on the correlation structure.

A small illustration. Figure 3 shows the ML AIC profile over a fine grid for three null trials of the base cell and three of the $t_{1/2} = 1$ week cell, with the CR2 t-statistic of the interaction across the same grid (verified; 40 seconds for the six profiles). The AIC profiles are shallow, often within 2 units of their minimum over a factor of four in the half-life, and the t-statistic changes smoothly with $t^a_{1/2}$, by up to 1.1 over the grid in one trial. The dashed profiles are those of F1 without the interaction term, the proposed F1A0 rule (Section 6.3); in these trials they are nearly identical to F1's, and on 50 further null trials per cell the two rules chose the same grid value in 86% and 88% of trials (verified, `56-aic-selection-summary.csv`; 164 seconds). This is an illustration on a few trials, not a size estimate.

6.2 Why a fixed half-life is right-sized even when it is wrong

Under the null with a purely predictive biomarker, the biomarker has no association with Y that depends on exposure, so the true slope vector has no component along $B_i^c D_{bc,it}$ for *any* regressor $D_{bc,it}$ chosen without looking at the outcome, and the interaction's limit is zero whatever $t^a_{1/2}$ is (derived; compendium, eq:bp-plim). A wrong half-life mis-scales the alternative, so it costs power; it does not create false positives. The stress test agrees: F1 at the fixed 0.5 week rejected 0.049 when the true half-life was 0, 0.061 when it was 1 week, 0.046 and 0.045 under Weibull decay, and at most 0.062 in any cell (verified, Table 8). With a prognostic biomarker the leak factors of Section 3.3 depend mildly on $t^a_{1/2}$, through $D_{bc,it}$ in the fitted slopes (derived), but no stress-test cell combined a prognostic pathway with a wrong half-life.

6.3 Status and intention

The project's position, set out in `docs/57`:

- **Validated now:** F1 with $t^a_{1/2}$ fixed in advance from pharmacology, with a sensitivity analysis over a grid of half-lives.

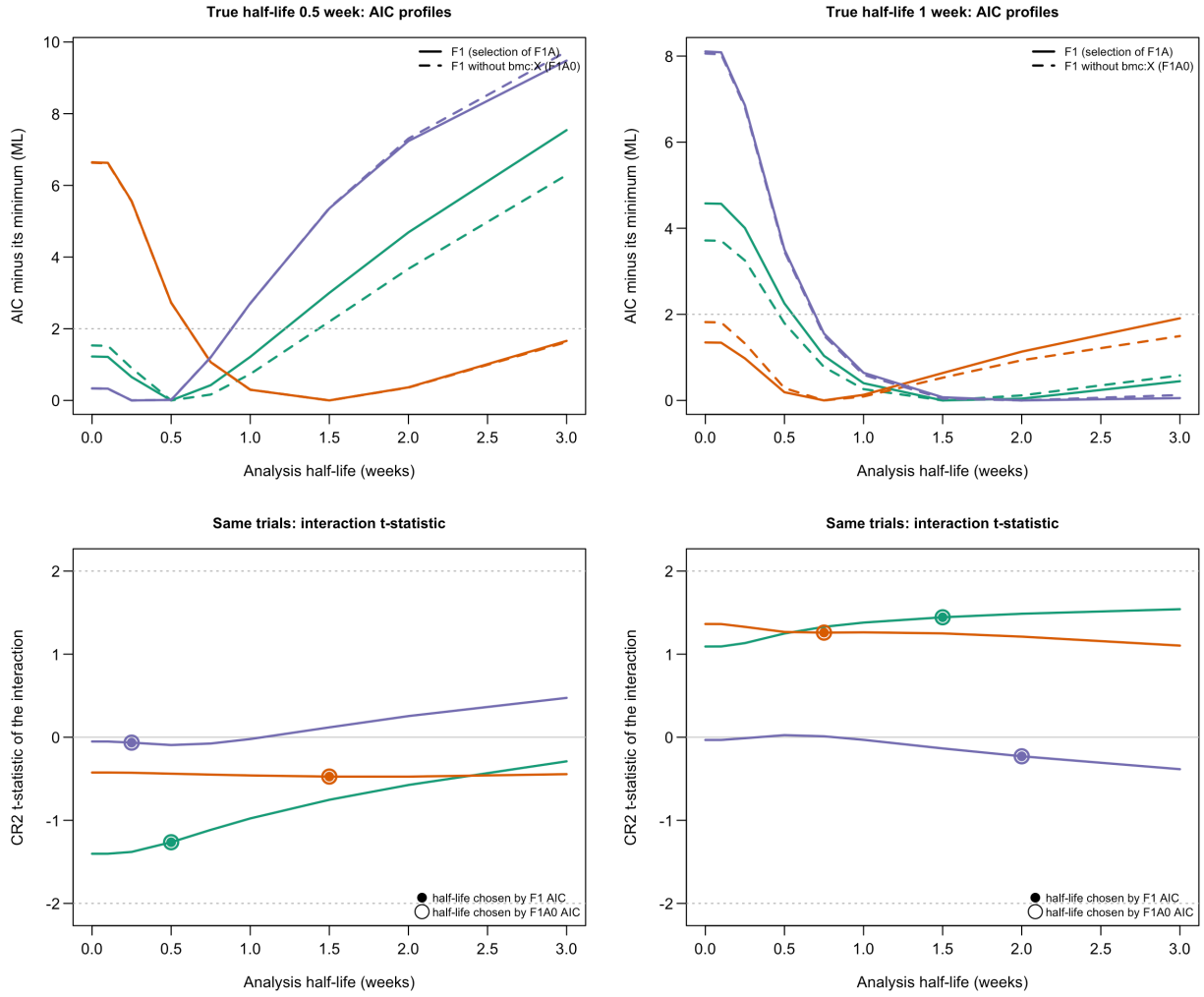


Figure 3: AIC selection of the analysis half-life on six null trials. Top: ML AIC minus its minimum over the half-life grid, for F1 with the interaction (solid; the F1A rule) and without it (dashed; the proposed F1A0 rule), true half-life 0.5 week (left) and 1 week (right). The dotted line marks 2 AIC units. Bottom: the CR2 t-statistic of the interaction in the same trials across the grid; filled circles mark the half-life chosen by the F1 AIC, open circles that chosen by the F1A0 AIC. Colors identify trials.

- **Intended, pending evaluation:** F1 with the half-life chosen by the data, because a reader can fairly ask why the analysis should use a half-life the data contradict. Two procedures are proposed and not yet run. **F1A0** chooses $t^a_{1/2}$ by AIC on F1 *without* the interaction term, where carryover is identified from the drug-response trajectory, then refits with the interaction and tests with CR2; it is approximately, not exactly, independent of the interaction test under the null (argued). **F1AB** calibrates the selection-then-test procedure by a parametric bootstrap from the fitted null model, repeating the selection in every resample; its operating characteristics would be evaluated with the warp-speed method of Giacomini, Politis and White [7] (proposal).
- **Report the half-life estimate in either case,** with a profile-likelihood interval, and state that a Hybrid trial carries little information about it (Section 6.1).

7. Alternatives and when to prefer them

Table 11: Alternatives to F1 (verified from the stored stress-test outputs; prognostic column: the largest of the four prognostic null cells)

Analysis	Prefer it when	Size, pooled / worst prognostic cell	Power, AR(1) / CS reference cell	Caveat
F1	the biomarker may be prognostic (the default)	0.051 / 0.047	0.364 / 0.522	approximately unbiased under a constant prognostic pathway; about a third less power than S under AR(1) data
G1, G2	as F1, with phase-specific residual variances; effectively equivalent	0.049 / 0.055; 0.047 / 0.062	0.354 / 0.546; 0.380 / 0.528	two more covariance parameters; G2's model-based inference relies on the working covariance
ES	the prognostic pathway, if any, is constant after baseline; under CS-like data	0.048 / 0.063	0.364 / 0.662	leaks an expectancy-scaled slope; loses most power when the prognostic slope grows (0.192)
S	independent evidence shows the biomarker is not prognostic (or only through baseline severity)	0.048 / 0.100	0.514 / 0.554	doubles its false-positive rate under a natural-course or placebo pathway
R (published type)	not recommended	0.069 / 0.159	0.398 / 0.596	liberal in standard cells and prognostic cells
F1A	not recommended	0.057 / 0.059	0.338 / 0.446	fails size through post-selection inference (Section 6)

Three remarks. First, the selection of F1 over G2 was a tie broken by the smaller number of covariance parameters (`docs/47`, Section 7); if all alternative cells, prognostic ones included, govern the selection, G1 and G2 lead F1 by about 0.05 in largest shortfall (quoted, `docs/47`, Section 7). Either is a defensible primary analysis. Second, report S as a sensitivity analysis: if F1 and S agree, the conclusion does not depend on the constancy assumption; if they disagree materially, the disagreement is itself evidence of a prognostic association (`docs/49`, Section 6). Third, in a fully blinded crossover the expectancy is constant and the expectancy terms are unnecessary and unidentified (`docs/49`, Section 8).

8. Practical recipe

The steps below fit F1 to a long data set `dat` with one row per participant and visit, baseline included, and columns `ptID` (participant), `t` (weeks from baseline, 0 at baseline), `Sx` (score), `bm` (biomarker), `on_drug` (logical), `tsd` (weeks since discontinuation: 0 on drug and before first exposure, cumulative within an off-drug run) and `De` (expectancy: 0 at baseline, 1 open label, 0.5 blinded).

1. **Fix $t^a_{1/2}$ before unblinding, from pharmacology.** For prazosin the stress test used 0.5 week. Write down the grid for the sensitivity analysis (for example 0, 0.25, 0.5, 1 and 2 weeks).
2. **Build the exposure indicator with the pre-exposure guard**, and check that no participant who starts on placebo is coded exposed.
3. **Center the biomarker** at its baseline mean. Do not dichotomize it.
4. **Fit the mean model with the expectancy terms** and the working covariance, by REML.
5. **Test with CR2 and Satterthwaite degrees of freedom**, clustered on participant, two-sided at 0.05.
6. **Report** the estimate per unit (and per SD) of the biomarker, its CR2 standard error, degrees of freedom, p-value and 95% interval; the coefficient of `bmc:De` as an exploratory measure of a prognostic or expectancy-linked association; S and ES as sensitivity analyses; F1 over the half-life grid; and the data's own estimate of the half-life (the ML AIC choice of F1 without the interaction term) with the statement that it is weakly identified. State the serial-correlation assumption behind the sample size.

```
library(nlme)
library(clubSandwich)

exposure_decay <- function(on_drug, tsd, t_half) {
  if (t_half <= 0) return(as.numeric(on_drug))
  ## tsd is 0 before first exposure too; without the guard a
  ## placebo-first participant is coded fully exposed
  ifelse(on_drug, 1, ifelse(tsd > 0, 0.5^(tsd / t_half), 0))
}

t_half_a <- 0.5
dat <- dat |>
  transform(bmc = bm - mean(bm[t == 0]),
            Dbc = exposure_decay(on_drug, tsd, t_half_a))
stopifnot(all(dat$Dbc[!dat$on_drug & dat$tsd == 0] == 0))

fit <- lme(Sx ~ bmc + t + Dbc + bmc:Dbc + De + bmc:De,
           random = ~ 1 | ptID,
           correlation = corCAR1(form = ~ t | ptID),
           data = dat, method = 'REML',
           control = lmeControl(opt = 'optim', maxIter = 200,
                                msMaxIter = 200))
coef_test(fit, vcov = 'CR2', test = 'Satterthwaite') |>
  subset(Coef == 'bmc:Dbc')
conf_int(fit, vcov = 'CR2', level = 0.95) |>
  subset(Coef == 'bmc:Dbc')
```

Applied to the $c_{bm} = 0.30$ trial of Section 4, this returns $\hat{\beta}_{bm:D} = -0.224$, SE 0.0877, $\nu = 25.8$, $p = 0.0167$ and the interval $(-0.405, -0.044)$, identical to the driver's F1 (verified; `56-recipe` check described in Section 10).

With the patched pmsimstats. On `rchendrickson/pmsimstats` at `58b32a9` with `analysis/patches/orig-to-refined/` applied, the same analysis is the default of `lme_analysis()`: `op$analysis = 'F1'`, `op$t_half_a = 0.5`,

and the result carries the estimate, the CR2 standard error, the p-value and the degrees of freedom (inspected; patch README, analysis table; verified there identical to the driver on five data sets).

Do not: choose $t^a_{1/2}$ by AIC with the interaction in the model and test as if fixed (F1A); use model-based standard errors with a compound-symmetry working covariance (R); drop the guard in $D_{bc,it}$; or plan the sample size for S and analyze with F1.

9. Limitations

- **Approximate unbiasedness.** F1 removes an expectancy-scaled prognostic slope exactly but a constant one only approximately in the Hybrid design, with residual leak factors of -0.02 to 0.17 of the prognostic slope depending on the working covariance (Section 3.3). The stress test simulated prognostic couplings of 0.3 only, one pathway at a time; larger couplings, combined pathways and a prognostic pathway combined with a wrong half-life were not run.
- **The stored results predate the refined construct.** They used the published recursive carryover and a fixed allocation; the rerun under anchored carryover and randomized allocation is pending. Anchored and recursive means differ by up to 0.48 points at $t_{1/2} = 0.5$ week in the Hybrid design (quoted; patch README), and the published-code analysis was insensitive to that change (`docs/53`, Section 8.3), but the stress-test rates have not been recomputed.
- **One design family, one effect size.** Only the Hybrid design was stressed, with the alternative at $c_{bm} = 0.30$ (0.25 under compound symmetry), and N from 35 to 150.
- **Gaussian outcome.** The construct is multivariate normal; skewed or bounded scores, and a biomarker acting on the rate rather than the size of response, are outside it.
- **Exploratory selection.** The study's criteria and analyses were revised as results came in (`docs/46`, Amendments 1 and 2), so F1 is a recommendation to be confirmed in an independent simulation.
- **The worked examples are single trials.** Section 4's $N = 70$ numbers illustrate the computation; the claims about bias rest on the large trial and the stored Monte Carlo results.
- **The data-chosen half-life is not yet evaluated** (Section 6.3, `docs/57`).

10. Reproducibility

Every number labeled verified in Sections 3 to 8 was computed by the scripts below, in `analysis/scripts/recommended-analysis/`, run from the repository root with R 4.6, `nlme` 3.1-169 and `clubSandwich` 0.7.0 (verified, host session; the container was not used).

- `56-load-driver.R`: loads the data-generating process and analyses of the stress-test driver (`analysis/scripts/quick-sim/ carryover-closed-form/ 11-strawman-stress-test.R`) by evaluating its top-level definitions only, with `--carryover recursive --allocation fixed` semantics, so that every fit is the driver's own code.
- `56-example-trial.R`: Table 3 and the crossover guard, the design weights and the drug-expectancy correlation, Tables 5 to 7 and Figure 1 (about 3 minutes, most of it the $N = 2000$ fits).
- `56-stress-summary.R`: Tables 8 to 10, the CR2 versus model-based comparison, κ, the F1A selection counts and Figure 2, from the stored files in `analysis/data/quick-sim/ carryover-closed-form/strawman-stress/` (`summary-full*.csv`, `replicates-full.rds`, `replicates-full-extra2.rds`, `decision-*-ds-nosym-all14.csv`); no new simulation (a few seconds).
- `56-aic-illustration.R`: Figure 3 and the F1A versus F1A0 selection comparison on 2 by 50 null trials (about 3.5 minutes).

For example, the command

```
Rscript analysis/scripts/recommended-analysis/56-example-trial.R
```

writes its tables to `analysis/data/recommended-analysis/` and the figure to `docs/figures/`. The recipe of Section 8 was run on the $c_{bm} = 0.30$ trial of Section 4 and matched the driver's F1 to all printed digits (verified, a scratch check not kept in the repository).

11. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013.
2. Hendrickson RC. pmsimstats: R package for simulation of aggregated N-of-1 trials. GitHub repository rchendrickson/pmsimstats, commit 58b32a92b99a8beb145de00c7bd8a0880f06f0f0 (2020-11-19). https://github.com/rchendrickson/pmsimstats.
3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*. 2002;28(2):169-181. (No DOI; published by Statistics Canada.)
4. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics*. 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004.
5. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
6. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558.
7. Giacomini R, Politis DN, White H. A warp-speed method for conducting Monte Carlo experiments involving bootstrap estimators. *Econometric Theory*. 2013;29(3):567-589. doi:10.1017/S0266466612000655.
8. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine*. 2009;28(20):2509-2530. doi:10.1002/sim.3639.
9. Senn S. *Cross-over Trials in Clinical Research*. 2nd ed. Chichester: Wiley; 2002. doi:10.1002/0470854596.
10. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086.
11. pmsimstats team. `docs/41`, `docs/42`, `docs/43`, `docs/45`, `docs/46`, `docs/47`, `docs/48`, `docs/49`, `docs/53`, `docs/57`; `analysis/report/ compendium-mathematics.tex`; report 01 (`analysis/report/ 01-dgp-mean-moderation-vs-mvn/report.Rmd`).